Class-Incremental Learning via LoRA-based Elastic Ensemble of Experts

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Abstract

Large-scale pre-trained vision models provide powerful representations for class-incremental learning, yet continuously adapting them to new classes without historical samples, task identifiers, or unrestricted parameter growth remains a fundamental challenge. Existing LoRA-based approaches typically follow either a one-adapter-per-task expansion paradigm or a fixed parameter-sharing strategy. The former leads to linear parameter growth as the task sequence expands, while the latter often induces severe cross-task interference. To address these limitations, we propose Elastic Ensemble of Experts ($\mathbf{E}^3$), a parameter-efficient framework for class-incremental learning. $\mathbf{E}^3$ treats LoRA modules as elastic experts whose task capacity is dynamically adjusted according to representational capacity saturation. Within each expert, we further introduce a novel parameter-partitioning mechanism that allocates LoRA parameters to disjoint, task-specific subspaces via magnitude-based importance estimation, thereby preventing parameter overwriting without requiring additional gradient-sensitivity computation. Moreover, $\mathbf{E}^3$ preserves and updates hierarchical covariance structures to mitigate interference at both the prediction level and the representation level and to provide a reference for capacity saturation detection. Extensive experiments on standard class-incremental learning benchmarks demonstrate that $\mathbf{E}^3$ achieves state-of-the-art or competitive performance, especially in long-horizon task sequences, while maintaining a favorable balance between stability, plasticity, and parameter efficiency. The code is available at https://github.com/yrluestc/EEE.

1 Introduction

The widespread adoption of large-scale pre-trained vision models has fundamentally transformed representation learning across diverse downstream applications. However, deploying these models in dynamic, non-stationary environments necessitates continuous adaptation to data without access to historical samples, a setting formalized as **class-incremental learning (CIL)**. In CIL, a model must sequentially learn novel classes while preserving knowledge of previously encountered ones, all without task identifiers at inference time. This paradigm inherently faces the *stability-plasticity dilemma*: aggressive adaptation to new data induces catastrophic forgetting, while excessive regularization stifles the acquisition of novel concepts.

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Recent advances in **parameter-efficient fine-tuning (PEFT)** have revitalized continual adaptation strategies. Low-Rank Adaptation (LoRA) [10], in particular, injects lightweight trainable matrices into frozen pre-trained backbones, drastically reducing compute and memory footprints. Building on this, several LoRA-based CIL approaches [27, 17] have proposed task-specific adapters or shared low-rank subspaces to mitigate forgetting. Nevertheless, these methods exhibit critical limitations in long-horizon streams. First, the prevalent "one-adapter-per-task" paradigm incurs linear parameter growth, violating strict efficiency constraints in resource-constrained deployments. Second, naive parameter sharing across tasks [14] leads to severe inter-task interference, as sequential gradient updates overwrite critical feature directions. Third, principled parameter importance metrics (e.g., Fisher information or gradient sensitivity) demand costly backward passes or Hessian approximations, rendering them impractical for continuous, low-budget scenarios. Consequently, a fundamental gap remains: how to dynamically allocate limited PEFT capacity across an open-ended task sequence while preserving historical knowledge without rehearsal or expensive sensitivity metrics?

To address this challenge, we introduce the **Elastic Ensemble of Experts ($\mathbf{E}^3$)**, a rehearsal-free, dynamically adaptive framework for parameter-efficient continual learning. Departing from rigid adapter expansions, $\mathbf{E}^3$ treats LoRA modules as elastic experts whose capacities are scheduled on-the-fly based on representational saturation. Our framework is organized around three synergistic roles: (i) **Dynamic Expert Capacity Scheduling** (§3.2 & §3.5), a macro-level capacity controller that automatically expands or contracts the number of tasks an expert can accommodate by monitoring subspace overlap, ensuring efficient resource utilization for orthogonal tasks while isolating highly confusable ones; (ii) **Group-Aware Parameter Partitioning** (§3.3), a micro-level parameter isolator that allocates disjoint parameter regions within each expert using magnitude-based importance, mitigating cross-task overwriting without gradient sensitivity computations; (iii) **Dual-Level Interference Mitigation** (§3.4), which preserves and updates hierarchical covariance structures to jointly mitigate interference at the prediction and representation levels and to provide a reference metric for capacity scheduling. Together, these components enable $\mathbf{E}^3$ to adaptively balance stability and plasticity while achieve parameter efficiency. Our main contributions are summarized as follows:

- We introduce a dynamic expert capacity scheduling policy that dynamically adjusts each expert's task capacity across sequential tasks, thereby breaking the rigid one-adapter-per-task paradigm and achieving state-of-the-art performance while preserving high parameter efficiency.
- We design Group-Aware Parameter Partitioning, a low-overhead, magnitude-driven allocation mechanism that enforces disjoint parameter spaces within shared experts, effectively isolating task-specific representations without expensive sensitivity analysis.
- We develop a hierarchical moment-based regularization objective for the classifier, coupled with orthogonal gradient projection for the LoRA expert, which jointly mitigate interference at the prediction and representation levels by utilizing compact covariance structures preserved from historical tasks.
- Extensive experiments on standard CL benchmarks demonstrate that our method outperforms existing PEFT-based and rehearsal-free baselines, with excellent parameter efficiency.

2 Related Work

Continual learning (CL) aims to acquire knowledge from non-stationary data streams without catastrophic forgetting [22]. Existing CL approaches are broadly categorized into three paradigms. (1) *Regularization-based methods* [11, 13, 30] impose constraints on parameter updates or distill historical knowledge via output/logit matching, but often suffer from rigidity in highly divergent task sequences. (2) *Replay-based methods* [18, 2, 28] store a small buffer of historical samples or generate pseudo-samples to consolidate memory, achieving strong performance at the cost of strict memory budgets and potential privacy concerns. (3) *Architecture-based methods* [25, 14, 29] dynamically expand networks or isolate task-specific subnetworks (prompts, adapters or LoRAs), inherently mitigating interference but frequently incurring unbounded parameter growth. Despite significant progress, most paradigms struggle to balance stability-plasticity performance, parameter efficiency, and task-agnostic inference under resource-constrained scenarios.

Low-Rank Adaptation-based CL has emerged as a prevalent strategy due to its minimal parameter overhead and compatibility with frozen backbones. Early works simply instantiate independent LoRA modules per task [4], leading to linear memory growth. Subsequent studies explore parameter sharing via low-rank merging [17] or rank reduction [27]. While these methods aim to mitigate the linear

parameter growth, they either rely on gradient-based inference for merge coefficients or depend on manually tuned heuristics, and do not fundamentally address the challenge of creating new parameters for each task. Our work departs from prior schemes that are either rigid or computationally intensive.

3 Methodology

3.1 Preliminaries

Problem Formulation. We consider a standard class-incremental learning (CIL) setting where data arrives as a sequence of N disjoint tasks $\{\mathcal{T}_1, \ldots, \mathcal{T}_N\}$. Each task $\mathcal{T}_t$ is associated with a training subset $\mathcal{D}_t = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{n_t}$ containing samples from a set of newly introduced classes. During the training phase of $\mathcal{T}_t$, the model accesses only $\mathcal{D}_t$, while all preceding datasets $\{\mathcal{D}_1, \ldots, \mathcal{D}_{t-1}\}$ are strictly unavailable. At inference time, the model must classify samples from the cumulative set $\bigcup_{j=1}^{t} \mathcal{D}_j$ without access to task identifiers. The core challenge lies in maintaining **stability** on historical classes while preserving **plasticity** for new classes, all under strict parameter budgets (**efficiency**) to prevent catastrophic forgetting.

Low-Rank Adaptation (LoRA). Parameter-efficient fine-tuning via LoRA [10] has emerged as a promising strategy for CIL [14, 27], as it minimizes trainable parameters while preserving the rich representational capacity of a pre-trained backbone. Consider an L-layer Vision Transformer (ViT) $F_\Omega : \mathcal{X} \to \mathcal{Z}$ with frozen pre-trained weights $\Omega = \{\mathbf{W}_0^{(\ell)}\}_{\ell=1}^{L}$. For the ℓ-th linear projection $\mathbf{W}_0^{(\ell)} \in \mathbb{R}^{m \times n}$, LoRA constrains the update $\Delta\mathbf{W}^{(\ell)}$ to a low-rank factorization:

$$\Delta\mathbf{W}^{(\ell)} = \mathbf{A}^{(\ell)}\mathbf{B}^{(\ell)}, \tag{1}$$

where $\mathbf{A}^{(\ell)} \in \mathbb{R}^{m \times r}$ and $\mathbf{B}^{(\ell)} \in \mathbb{R}^{r \times n}$ are trainable down- and up-projection matrices, respectively, with rank $r \ll \min\{m, n\}$ controlling the adaptation capacity. The forward pass is consequently modified as:

$$\mathbf{z}^{(\ell)} = \mathbf{W}_0^{(\ell)}\mathbf{x}^{(\ell)} + \mathbf{A}^{(\ell)}\mathbf{B}^{(\ell)}\mathbf{x}^{(\ell)}, \tag{2}$$

where $\mathbf{x}^{(\ell)}$ denotes the input activation to ℓ-th layer, and $\mathbf{z}^{(\ell)}$ is the corresponding output. Throughout training, $\mathbf{W}_0^{(\ell)}$ remains fixed, ensuring that only $\{\mathbf{A}^{(\ell)}, \mathbf{B}^{(\ell)}\}_{\ell=1}^{L}$ are optimized.

3.2 Dynamic Expert Capacity Scheduling

Let $\mathcal{E} = (\mathcal{E}_1, \mathcal{E}_2, \ldots, \mathcal{E}_K)$ be the sequence of experts created during training, where $K \leq N$ is determined dynamically. Each expert $\mathcal{E}_k$ has a predefined capacity $\mathcal{C}_k \in \mathbb{Z}^+$ and is designed to handle a series of consecutive tasks whose indices lie in $[t_k^{\text{start}}, t_k^{\text{end}}]$, where $t_k^{\text{end}} = t_k^{\text{start}} + \mathcal{C}_k - 1$. Accordingly, once the capacity of $\mathcal{E}_k$ is exhausted, its parameters are frozen and a new expert is created to adapt to subsequent tasks. However, *how aggressively should an expert be reused?* Intuitively, if an expert successfully completes its assigned tasks, this indicates sufficient representational flexibility under the current capacity setting, allowing the next expert to be assigned a larger capacity; conversely, if an expert becomes saturated, this suggests that subsequent experts should be more conservative, i.e., assigned smaller capacities, to prevent overloading. This offers a flexible trade-off between parameter efficiency and interference avoidance. We implement this capacity scheduling process in four steps:

(1) **Initialization.** During the initial learning phase, we create the first expert with minimal capacity $\mathcal{C}_1 = 1$, aiming to enable stable adaptation to the first task without cross-task interference.

(2) **Progressive Expansion.** If a certain expert $\mathcal{E}_k$ successfully completes its assigned tasks without saturation, we incrementally probe the capacity boundary for the next expert:

$$\mathcal{C}_{k+1} \leftarrow \mathcal{C}_k + 1, \tag{3}$$

where the increment reflects growing confidence in the expert's representational robustness.

(3) **Adaptive Fallback.** If saturation is detected after training a task (detailed in § 3.5), we immediately freeze $\mathcal{E}_k$ and reduce the capacity for the subsequent expert to prevent overloading:

$$\mathcal{C}_{k+1} \leftarrow \max(\mathcal{C}_k - 1, \mathcal{C}_{\min}), \tag{4}$$

with $\mathcal{C}_{\min} = 1$ enforcing a lower bound. A new expert $\mathcal{E}_{k+1}$ is then created to resume learning.

(4) **Termination.** The process repeats until all N tasks are processed. The final expert $\mathcal{E}_K$ may handle fewer than $\mathcal{C}_K$ tasks if the stream concludes early.

This policy ensures that experts automatically specialize for highly overlapping tasks (via reduced capacity) while efficiently sharing resources for orthogonal or simple tasks (via expanded capacity).

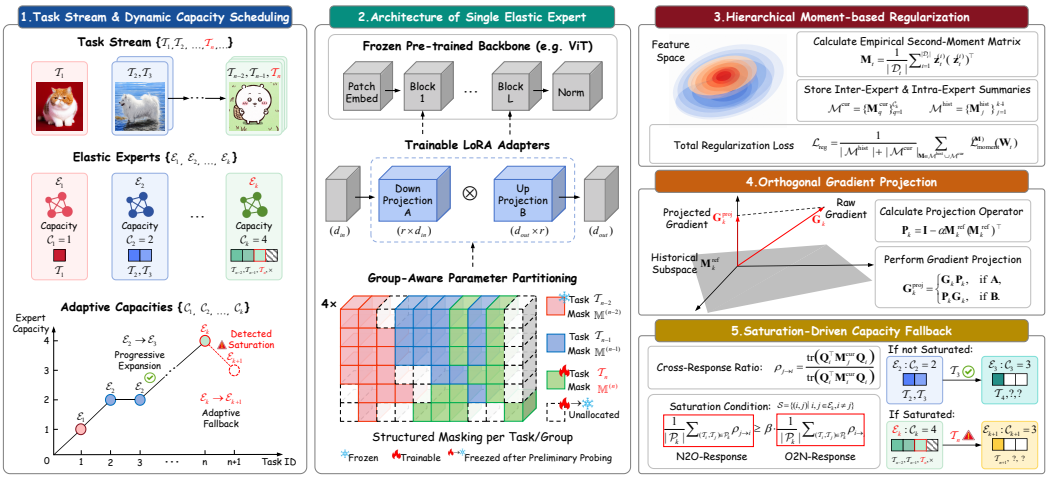


Figure 1: Overview of the proposed $\mathbf{E}^3$ training framework. $\mathbf{E}^3$ processes a class-incremental task stream using an elastic ensemble of LoRA-based experts whose capacities are dynamically scheduled according to representational saturation. Each expert employs group-aware parameter partitioning to allocate disjoint task-specific LoRA subspaces via structured masks, preventing new-task updates from overwriting previously assigned parameters. Historical knowledge is further preserved by hierarchical moment-based regularization and orthogonal gradient projection, which reduces interference at both the prediction and representation levels.

3.3 Group-Aware Parameter Partitioning

Although dynamic expert capacity scheduling enables multiple tasks to share an expert, directly updating the same LoRA parameters across tasks inevitably induces interference. Feasible expert sharing therefore requires a parameter-level mechanism that allocates different tasks to distinct parameter subspaces within a shared expert. Gradient-based sensitivity [7] and Fisher information [11] provide principled importance estimates but incur additional computation, which is costly for long task streams. Parameter magnitude [15, 24, 6] offers a simple yet effective alternative: parameters that receive large updates during task adaptation are likely to encode task-relevant information. We thus adopt magnitude-based importance as a zero-overhead allocation signal, motivated by the need for a lightweight mechanism that can be repeatedly applied in incremental learning without substantial memory or computational overhead. Accordingly, we propose **Group-Aware Parameter Partitioning (GAPP)**, which partitions LoRA parameters using both structural priors and magnitude-based task importance to assign each task a fair and disjoint parameter subset, thereby preventing new-task learning from overwriting parameters assigned to earlier tasks.

Specifically, let us define $\Theta_{\text{LoRA}}^{(k)} = \{\mathbf{A}^{(\ell)}, \mathbf{B}^{(\ell)}\}_{\ell=1}^{L}$ as the trainable parameters of a certain expert $\mathcal{E}_k$. The proposed GAPP aims to partition $\Theta_{\text{LoRA}}^{(k)}$ into $\mathcal{C}_k$ disjoint subsets $\{\Theta_1^{(k)}, \ldots, \Theta_{\mathcal{C}_k}^{(k)}\}$ such that:

$$\bigcup_{j=1}^{\mathcal{C}_k} \Theta_j^{(k)} = \Theta_{\text{LoRA}}^{(k)}, \quad \Theta_i^{(k)} \cap \Theta_j^{(k)} = \emptyset \ \ \forall i \neq j, \tag{5}$$

where each subset receives an equal parameter budget $|\Theta_i^{(k)}| = |\Theta_j^{(k)}|$ to prevent capacity bias, where $|\Theta_i^{(k)}|$ denotes the total number of parameters within $\Theta_i^{(k)}$. The proposed partitioning process consists of a two-stage protocol, described in the following.

Stage 1: Preliminary Probing. All currently unallocated parameters $\Theta_{\text{free}} \subseteq \Theta_{\text{LoRA}}^{(k)}$ are trained for σ epochs on the current task $\mathcal{T}_t$. Here, σ is a fixed hyperparameter chosen to allow near-convergence. This short probing phase allows the optimization dynamics to reveal which parameters are most useful for the task. Importantly, parameters already assigned to previous tasks remain unavailable, so probing does not overwrite historical allocations.

Stage 2: Structured Masking. To account for heterogeneous roles of LoRA components, we partition $\Theta_{\text{LoRA}}^{(k)}$ into four structural groups $\mathcal{G} = \{g_{\text{down}}^{\text{shallow}}, g_{\text{down}}^{\text{deep}}, g_{\text{up}}^{\text{shallow}}, g_{\text{up}}^{\text{deep}}\}$ via a mapping $\phi : \Theta_{\text{LoRA}}^{(k)} \to \mathcal{G}$.

The depth split occurs at layer $\lfloor L/2 \rfloor$, separating early and late representations. For each parameter θ, let $s(\theta) = |\theta|$ be its raw magnitude. Within each group g, we compute a group-specific threshold at the ρ-th percentile by:

$$\tau_g = Q_\rho\big(\{s(\theta) : \phi(\theta) = g\}\big), \tag{6}$$

where $\rho = 1 - \frac{1}{\mathcal{C}_k}$ and $Q_\rho(\cdot)$ returns the empirical ρ-quantile. The selection mask for task $\mathcal{T}_t$ is then defined by:

$$\mathbb{M}^{(t)}(\theta) = \mathbb{I}\big(s(\theta) \geq \tau_{\phi(\theta)}\big), \tag{7}$$

where $\mathbb{I}(\cdot)$ is the indicator function. To preserve historical allocations, we aggregate past masks and construct a freeze mask:

$$\mathbb{M}^{\text{union}} = \bigvee_{t'<t} \mathbb{M}^{(t')}, \quad \mathbb{M}^{\text{freeze}} = \mathbf{1} - \mathbb{M}^{\text{union}}, \tag{8}$$

where $\mathbb{M}^{\text{freeze}}(\theta) = 1$ indicates that θ remains trainable for the current task. During preliminary training, importance scores are computed exclusively on available parameters:

$$s^{\text{masked}}(\theta) = s(\theta) \odot \mathbb{M}^{\text{freeze}}(\theta), \tag{9}$$

where $\odot$ denotes element-wise multiplication. The gradient update process is defined by:

$$\nabla\theta \leftarrow \nabla\theta \odot \mathbb{M}(\theta), \quad \mathbb{M}(\theta) = \begin{cases} \mathbb{M}^{\text{freeze}}(\theta), & \text{if } e \leq \sigma, \\ \mathbb{M}^{(t)}(\theta), & \text{if } e > \sigma, \end{cases} \tag{10}$$

where e denotes the current training epoch. This progressive allocation guarantees that once a parameter is assigned to a task, it remains fixed for subsequent tasks within the same expert, effectively reducing interference while preserving critical subspaces.

3.4 Dual-Level Interference Mitigation

Macro-level expert capacity scheduling and micro-level disjoint parameter allocation structurally reduce task interference. However, forgetting can still arise along two residual pathways that are not eliminated by structural isolation: (i) the *prediction level*, where the shared classification head assigns drifting logits to previously seen classes as new classes are appended, and (ii) the *representation level*, where LoRA updates for the current task drift along feature directions that were important for earlier tasks. We therefore complement the structural isolation with a dual-level mitigation scheme built on a single rehearsal-free statistic, the task moment matrix, which is shared by both levels but exploited differently at each. This common foundation keeps the two mechanisms coherent for long task streams.

Shared Moment Statistics. For each task $\mathcal{T}_t$, we characterize the feature distribution without storing raw samples by its empirical second-moment matrix:

$$\mathbf{M}_t = \frac{1}{|\mathcal{D}_t|} \sum\nolimits_{i=1}^{|\mathcal{D}_t|} \mathbf{z}_t^{(i)} (\mathbf{z}_t^{(i)})^\top, \tag{11}$$

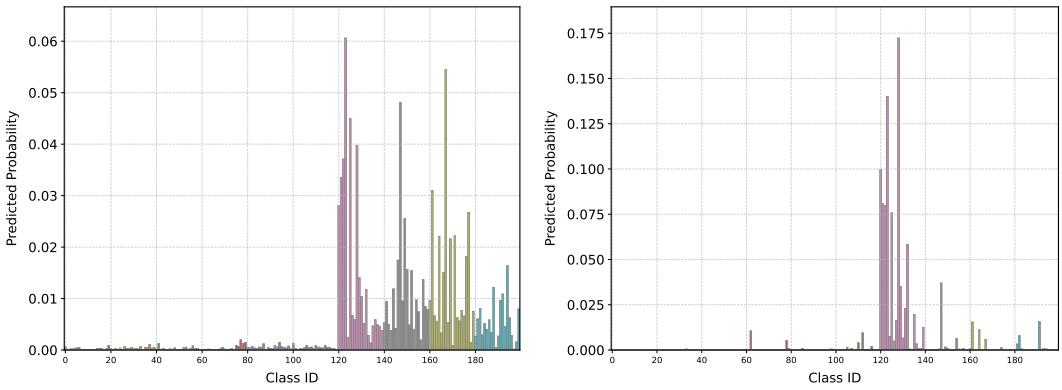
where $|\mathcal{D}_t|$ denotes the total number of data samples within $\mathcal{D}_t$ and the superscript T in $\mathbf{z}_t^{(i)}$ denotes the matrix transpose. $\mathbf{z}_t^{(i)}$ is the ℓ_2-normalized feature of the i-th sample. $\mathbf{M}_t$ compactly encodes the covariance structure of task-specific features, serving as a rehearsal-free distributional proxy for both levels below.

Prediction-Level Mitigation via Hierarchical Moment-based Regularization. At this level, the objective is to prevent newly introduced class weights from occupying directions that encode historical classes. We therefore regularize the classifier against the feature directions summarized by the moment statistics. Since the dynamic expert capacity scheduling organizes tasks into experts, we track the historical feature structure at two granularities [21, 3]. One is an intra-expert memory:

$$\mathcal{M}^{\text{cur}} = \{\mathbf{M}_q^{\text{cur}}\}_{q=1}^{\mathcal{C}_k}, \tag{12}$$

which stores fine-grained moments for tasks currently assigned to expert $\mathcal{E}_k$. The second is an inter-expert memory:

$$\mathcal{M}^{\text{hist}} = \{\mathbf{M}_j^{\text{hist}}\}_{j=1}^{k-1}, \tag{13}$$



(a) Predicted probability distribution with frozen *linear* classifier.

(b) Predicted probability distribution with frozen *cosine* classifier.

Figure 2: A comparison of predicted probability distributions on Task 7 between the linear and cosine classifiers, after training on all 10 tasks of ImageNet-A. When each task is completed, the classifier weights (and bias, if any) corresponding to that task are frozen permanently.

which summarizes knowledge from previously archived experts $\{\mathcal{E}_1, \cdots, \mathcal{E}_{k-1}\}$. When $\mathcal{E}_k$ completes its capacity, the current moment set is aggregated into the historical pool:

$$\mathbf{M}_k^{\text{hist}} = \frac{1}{|\mathcal{M}^{\text{cur}}|} \sum\nolimits_{\mathbf{M} \in \mathcal{M}^{\text{cur}}} \mathbf{M}, \tag{14}$$

and $\mathcal{M}^{\text{cur}}$ is cleared for the next expert. $|\mathcal{M}^{\text{cur}}|$ denotes the total number of elements within $\mathcal{M}^{\text{cur}}$. This hierarchy allows the model to preserve fine-grained task information within an active expert while maintaining compact summaries for archived experts.

Before perform regularization, we additionally replace the linear logit with a cosine-normalized one. For a feature embedding $\mathbf{z} \in \mathbb{R}^D$ and class weights $\{\mathbf{w}_c\}_{c=1}^{C_t}$ corresponding to task $\mathcal{T}_t$, the logit is

$$\text{logit}_c = \frac{\mathbf{w}_c^\top \mathbf{z}}{\|\mathbf{w}_c\|_2 \cdot \|\mathbf{z}\|_2}, \tag{15}$$

where $\|\cdot\|_2$ is the ℓ_2 norm. As shown in Fig. 2, this mitigates magnitude-induced miscalibration for frozen historical classes.

Then, we introduce a moment-based regularization term to enforce a historical directional constraint. Let $\widetilde{\mathbf{W}}_t = \{\widetilde{\mathbf{w}}_c\}_{c=1}^{C_t}$ denote the column-wise ℓ_2-normalized classifier weights, where $\widetilde{\mathbf{w}}_c = \mathbf{w}_c / \|\mathbf{w}_c\|_2$. The alignment penalty for a single moment matrix $\mathbf{M}$ is:

$$\mathcal{L}_{\text{moment}}^{(\mathbf{M})}(\mathbf{W}_t) = \left\|\widetilde{\mathbf{W}}_t \mathbf{M}^{1/2}\right\|_F^2 = \text{tr}\big(\widetilde{\mathbf{W}}_t \mathbf{M} \widetilde{\mathbf{W}}_t^\top\big), \tag{16}$$

which penalizes weight vectors aligning with high-variance historical directions. In practice, this penalty is applied to the classifier weights associated with *newly* introduced classes, while historical class weights remain frozen. The total regularization loss averages across all tracked moments:

$$\mathcal{L}_{\text{reg}} = \frac{1}{|\mathcal{M}^{\text{hist}}| + |\mathcal{M}^{\text{cur}}|} \sum\nolimits_{\mathbf{M} \in \mathcal{M}^{\text{hist}} \cup \mathcal{M}^{\text{cur}}} \mathcal{L}_{\text{moment}}^{(\mathbf{M})}(\mathbf{W}_t). \tag{17}$$

Representation-Level Mitigation via Orthogonal Gradient Projection. Although the regularization term above constrains the classifier, feature extractor gradients may still overwrite previously learned representations. To complement it at the representation level, we project LoRA gradients onto the orthogonal complement of the historical feature subspace within $\mathcal{E}_k$. Let us define $\mathbf{M}_k^{\text{ref}} = \frac{1}{|\mathcal{M}^{\text{cur}}|} \sum_{\mathbf{M} \in \mathcal{M}^{\text{cur}}} \mathbf{M}$ as the current expert's reference matrix. We construct a projection operator as:

$$\mathbf{P}_k = \mathbf{I} - \alpha \mathbf{M}_k^{\text{ref}} (\mathbf{M}_k^{\text{ref}})^\top, \tag{18}$$

where $\alpha \in (0, 1]$ controls projection strength, and $\mathbf{I}$ is the identity matrix. Geometrically, $\mathbf{P}_k$ removes gradient components aligned with dominant historical directions in $\mathbf{M}_k^{\text{ref}}$. For a gradient $\mathbf{G}_k$ of LoRA expert $\mathcal{E}_k$, projection is applied along the feature dimension. If the feature dimension corresponds

Table 1: Performance comparison with CL methods on ImageNet-R across different task lengths. The best results are highlighted in bold, and the second-best results are underlined.

Method	ImageNet-R 5 tasks		ImageNet-R 10 tasks		ImageNet-R 20 tasks	
	Acc↑	AAA↑	Acc↑	AAA↑	Acc↑	AAA↑
Full Fine-Tuning	$64.92_{(0.87)}$	$75.57_{(0.50)}$	$60.57_{(1.06)}$	$72.31_{(1.09)}$	$49.95_{(1.31)}$	$65.32_{(0.84)}$
L2P[26]	$73.04_{(0.71)}$	$76.94_{(0.41)}$	$71.26_{(0.44)}$	$76.13_{(0.46)}$	$68.97_{(0.51)}$	$74.16_{(0.32)}$
DualPrompt[25]	$69.99_{(0.57)}$	$72.24_{(0.41)}$	$68.22_{(0.20)}$	$73.81_{(0.39)}$	$65.23_{(0.45)}$	$71.30_{(0.16)}$
CODA-Prompt[19]	$76.63_{(0.27)}$	$80.30_{(0.28)}$	$74.05_{(0.41)}$	$78.14_{(0.39)}$	$69.38_{(0.33)}$	$73.95_{(0.63)}$
HiDe-Prompt[21]	$74.77_{(0.25)}$	$78.15_{(0.24)}$	$74.65_{(0.14)}$	$78.46_{(0.18)}$	$73.59_{(0.19)}$	$77.93_{(0.19)}$
InfLoRA[14]	$76.95_{(0.23)}$	$81.81_{(0.14)}$	$74.75_{(0.64)}$	$80.67_{(0.55)}$	$69.89_{(0.56)}$	$76.68_{(0.57)}$
SD-LoRA[27]	$79.15_{(0.20)}$	$83.01_{(0.42)}$	$77.34_{(0.35)}$	$82.04_{(0.24)}$	$75.26_{(0.37)}$	$80.22_{(0.72)}$
LoRA-P&M[17]	$81.47_{(0.56)}$	$85.96_{(0.52)}$	$79.95_{(0.18)}$	$85.29_{(0.93)}$	$76.37_{(0.09)}$	$82.77_{(0.71)}$
$\mathbf{E}^3$	$\underline{85.89}_{(0.29)}$	$\underline{86.97}_{(0.46)}$	$\underline{85.45}_{(0.53)}$	$\underline{88.80}_{(0.24)}$	$\underline{82.76}_{(0.31)}$	$\underline{86.43}_{(0.42)}$
$\mathbf{E}^3$ (Full Expansion)	$\mathbf{91.10}_{(0.27)}$	$\mathbf{91.10}_{(0.16)}$	$\mathbf{93.26}_{(0.55)}$	$\mathbf{93.59}_{(0.34)}$	$\mathbf{94.20}_{(0.24)}$	$\mathbf{94.48}_{(0.43)}$

to columns, we right-multiply by $\mathbf{P}_k$ (i.e., applies to $\mathbf{A}$ matrices); if it corresponds to rows, we left-multiply by $\mathbf{P}_k$ (i.e., applies to $\mathbf{B}$ matrices):

$$\mathbf{G}_k^{\text{proj}} = \begin{cases} \mathbf{G}_k\mathbf{P}_k, & \text{if } \mathbf{P}_k \text{ applies to } \mathbf{A}, \\ \mathbf{P}_k\mathbf{G}_k, & \text{if } \mathbf{P}_k \text{ applies to } \mathbf{B}. \end{cases} \tag{19}$$

Unified Optimization Objective. The complete training objective for task $\mathcal{T}_t$ combines standard cross-entropy with our hierarchical regularization term:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}}, \tag{20}$$

where $\mathcal{L}_{\text{CE}}$ denotes the cross entropy loss calculated on the current task learning and λ_{reg} balances stability and plasticity. During backpropagation, gradients are first *masked* via Eq. (10) and then *projected* via Eq. (19) before the optimizer step.

Taken together, the two levels play complementary roles: hierarchical moment-based regularization prevents new-class weights from aligning with historical feature directions at the prediction level, while orthogonal gradient projection restricts updates to the experts, preventing the feature extractor itself from drifting along those directions at the representation level.

3.5 Saturation-Driven Capacity Fallback

A crucial question in dynamic expert capacity scheduling (§3.2) is when an expert should stop accepting new tasks. If an expert is reused too aggressively, tasks may compete for overlapping feature directions and degrade one another. If experts are created too frequently, the model loses parameter efficiency. We therefore introduce a saturation detection mechanism at the end of each task once the current expert contains at least two tasks, which empirically estimates whether the expert has reached its representational limit. The mechanism is based on the intuition that compatible tasks should occupy relatively distinct feature subspaces, whereas saturated experts exhibit strong cross-task activation in the same dominant directions. Let $\mathcal{S}_k$ denote the tasks handled by $\mathcal{E}_k$, ordered by arrival, and let $\mathbf{Q}_i \in \mathbb{R}^{D\times d}$ contain the top-d eigenvectors of $\mathbf{M}_i^{\text{cur}}$. For tasks $\mathcal{T}_i, \mathcal{T}_j \in \mathcal{S}_k$, we define the relative cross-response of source task $\mathcal{T}_j$ on the principal subspace of target task $\mathcal{T}_i$ as

$$\rho_{j\to i} = \frac{\text{tr}\big(\mathbf{Q}_i^\top \mathbf{M}_j^{\text{cur}} \mathbf{Q}_i\big)}{\text{tr}\big(\mathbf{Q}_i^\top \mathbf{M}_i^{\text{cur}} \mathbf{Q}_i\big)}, \tag{21}$$

where the denominator is the self-response of $\mathcal{T}_i$ on its own principal subspace. Thus, $\rho_{j\to i}$ measures how strongly the feature distribution of $\mathcal{T}_j$ activates the dominant directions of $\mathcal{T}_i$ relative to $\mathcal{T}_i$'s own activation.

Saturation condition: Because tasks arrive sequentially, the two directions of a task pair need not be symmetric. We therefore compare the average activation from newer tasks into older subspaces with the reverse average activation from older tasks into newer subspaces. Let $\mathcal{P}_k = \{(\mathcal{T}_i, \mathcal{T}_j) \mid \mathcal{T}_i, \mathcal{T}_j \in \mathcal{S}_k,\ i < j\}$ contain all chronologically ordered task pairs (i.e., $i < j$ means $\mathcal{T}_i$ arrives before $\mathcal{T}_j$). We

Table 2: Performance comparison on ImageNet-A and DomainNet.

Method	ImageNet-A 10 tasks		DomainNet 5 tasks	
	Acc ↑	AAA ↑	Acc ↑	AAA ↑
Full Fine-Tuning	$16.31_{(7.89)}$	$30.04_{(13.18)}$	$51.46_{(0.47)}$	$67.08_{(1.13)}$
L2P [26]	$42.94_{(1.27)}$	$51.40_{(1.95)}$	$70.26_{(0.25)}$	$75.83_{(0.98)}$
DualPrompt [25]	$45.49_{(0.96)}$	$54.68_{(1.24)}$	$68.26_{(0.90)}$	$73.84_{(0.45)}$
CODA-Prompt [19]	$45.36_{(0.78)}$	$57.03_{(0.94)}$	$70.58_{(0.53)}$	$76.68_{(0.44)}$
HiDe-Prompt [21]	$42.70_{(0.60)}$	$56.32_{(0.40)}$	$72.20_{(0.08)}$	$77.01_{(0.04)}$
InfLoRA [14]	$49.20_{(1.12)}$	$60.92_{(0.61)}$	$71.59_{(0.23)}$	$78.29_{(0.50)}$
SD-LoRA [27]	$55.96_{(0.73)}$	$64.95_{(1.63)}$	$72.82_{(0.37)}$	$78.89_{(0.50)}$
LoRA-P&M [17]	$56.57_{(0.78)}$	$65.35_{(1.81)}$	$\underline{76.91}_{(0.53)}$	$\underline{85.27}_{(0.07)}$
$\mathbf{E}^3$	$\underline{65.84}_{(0.65)}$	$\underline{74.11}_{(1.10)}$	$73.10_{(0.61)}$	$78.87_{(0.31)}$
$\mathbf{E}^3$ (Full Expansion)	$\mathbf{71.49}_{(0.55)}$	$\mathbf{77.19}_{(1.30)}$	$\mathbf{85.58}_{(0.28)}$	$\mathbf{86.56}_{(0.16)}$

declare $\mathcal{E}_k$ saturated when:

$$\frac{1}{|\mathcal{P}_k|}\sum\nolimits_{(\mathcal{T}_i,\mathcal{T}_j)\in\mathcal{P}_k}\rho_{j\to i} \geq \beta \cdot \frac{1}{|\mathcal{P}_k|}\sum\nolimits_{(\mathcal{T}_i,\mathcal{T}_j)\in\mathcal{P}_k}\rho_{i\to j}, \tag{22}$$

where the left-hand side averages the newer-to-older (N2O) responses $\rho_{j\to i}$ and the right-hand side averages the older-to-newer (O2N) responses $\rho_{i\to j}$. For well-separated tasks, newer tasks intrude little into the principal subspaces of older tasks, so $\rho_{j\to i}$ stays well below $\rho_{i\to j}$; saturation occurs when the newer-to-older activation reaches β times the reverse activation, indicating that the expert can no longer isolate the tasks it already contains. Upon saturation detection, we freeze $\mathcal{E}_k$, and instantiate $\mathcal{E}_{k+1}$ with reduced capacity via Eq. (4). Specifically, unallocated LoRA parameters in $\mathcal{E}_k$ (if any) are neither discarded nor transferred but are instead frozen, since the unstructured partitioning in GAPP (§3.3) cannot be directly pruned without breaking dimension alignment, and since freezing preserves the learned representations for inference while preventing future overwriting.

This mechanism guarantees that experts automatically specialize for highly confusable tasks while maximizing resource utilization for orthogonal ones. An analysis of the number of experts and trainable/total parameters during the learning process on each dataset is provided in Appendix C. The learning process of the proposed framework is shown in Fig. 1 and Algorithm 1. Notably, during inference, each test sample is processed by all existing experts to obtain their respective prediction logits. The output from the expert exhibiting the minimum predictive entropy (indicating the highest confidence) is selected as the final prediction. A detailed analysis of inference is provided in the Appendix B.

4 Experiments

4.1 Experimental Setups

Evaluation Benchmarks and Metrics. We base our experiments on [14, 27, 17] and evaluate $\mathbf{E}^3$ on five standard CL benchmarks: ImageNet-R (split into 5/10/20 tasks) [8], ImageNet-A (10 tasks) [9], DomainNet (5 tasks) [16], CIFAR100 (10 tasks) [12], and CUB200 (10 tasks) [20]. Two standard metrics are used: average accuracy (Acc), defined as the average final accuracy over all tasks, and average anytime accuracy (AAA), which is the average accuracy over all seen tasks calculated incrementally after learning each new task.

Competing Methods and Implementation Details. The proposed $\mathbf{E}^3$ is compared against SOTA ViT-based CL methods: L2P [26], DualPrompt [25], CODA-Prompt [19], HiDe-Prompt [21], InfLoRA [14], SD-LoRA [27], and LoRA-P&M [17], with "Full Fine-Tuning" and "Full Expansion" (one expert per task) as lower and upper performance bounds, respectively. We adopt a ViT-B/16 backbone [5] (pre-trained on ImageNet-21K then fine-tuned on ImageNet-1K) as in [27, 17]. LoRA modules (rank=10) are added to the query and value projections in all multi-head attention layers. Training uses SGD [1] with a learning rate of 0.03, a batch size of 32, and runs for 10 epochs per task. Results report the mean and standard deviation over five runs on a single NVIDIA RTX 3090 GPU.

4.2 Experimental Results

Results on Different CL Benchmarks. As shown in Tab. 1 and Tab. 7, $\mathbf{E}^3$ achieves state-of-the-art performance on most evaluated benchmarks under the ViT-B/16 backbone. On **ImageNet-R**,

Table 3: Ablation analysis of the proposed $\mathbf{E}^3$. Each experimental stage incorporates all modules from the preceding stages (e.g., Exp.3-1 corresponds to LoRA + Cosine* + GAPP + Grad Proj.).

Exp. No.	Variant	ImageNet-R 20 tasks		ImageNet-A 10 tasks	
		Acc↑	AAA↑	Acc↑	AAA↑
(1-1)	LoRA + Linear	41.31	47.79	27.24	40.91
(1-2)	LoRA + Linear*	50.35	64.06	41.42	56.55
(1-3)	LoRA + Cosine*	53.37	69.33	45.27	61.61
(2-1)	+ Partition	59.12	74.21	45.35	65.31
(2-2)	+ GAPP	74.49	82.02	52.02	68.10
(2-3)	+ GAPP + Grad Proj.	76.76	83.31	55.52	69.28
(3-1)	+ Capacity Fallback	80.70	85.78	57.34	70.29
(3-2)	+ Capacity Fallback + $\mathcal{L}_{\text{reg}(\mathcal{M}_{\text{cur}})}$	81.45	86.08	62.09	71.37
(3-3)	+ Capacity Fallback + $\mathcal{L}_{\text{reg}}$ (full $\mathbf{E}^3$)	**82.76**	**86.43**	**65.84**	**74.11**

Table 4: Comparison of different parameter importance metrics within GAPP (Based on Exp. 2-2). Diagonal Fisher and magnitude are broadly comparable, whereas random masking is substantially worse. Magnitude requires no additional backward pass or importance storage.

Metric	INR (5)		INR (10)		INR (20)		INA (10)	
	Acc↑	AAA↑	Acc↑	AAA↑	Acc↑	AAA↑	Acc↑	AAA↑
Random	64.08	77.19	73.30	79.23	45.49	62.51	46.56	61.67
Fisher	81.58	**84.01**	**81.49**	**86.67**	**76.22**	**82.59**	50.97	**69.33**
Magnitude (Ours)	**82.41**	83.93	80.00	82.78	74.49	82.02	**52.02**	68.10

$\mathbf{E}^3$ consistently outperforms all competing methods across varying task granularities. Specifically, with 20 sequential tasks, $\mathbf{E}^3$ surpasses the strongest baseline LoRA-P&M by **+6.39%** in average accuracy (Acc) and **+3.66%** in average incremental accuracy (AAA). Notably, the performance margin widens as the task sequence lengthens (from +4.42% Acc at 5 tasks to +6.39% at 20 tasks), demonstrating that our elastic capacity scheduling and interference-aware parameter allocation become increasingly beneficial in long-horizon continual learning scenarios. On the more challenging **ImageNet-A** benchmark, which emphasizes adversarial robustness and distribution shift, $\mathbf{E}^3$ exhibits particularly strong gains: it exceeds LoRA-P&M by **+9.27%** in Acc and **+8.76%** in AAA. This substantial improvement suggests that our hierarchical moment regularization and orthogonal gradient projection effectively preserve discriminative feature subspaces. Besides, performance comparisons on DomainNet demonstrate that $\mathbf{E}^3$ is also effectively applicable in Domain-Incremental Learning scenarios. Although it slightly underperforms LoRA-P&M, our method still achieves the best performance in the "Full Expansion" version with the same number of LoRA modules.

Ablation Studies. Tab. 3 details a stepwise ablation of $\mathbf{E}^3$. The asterisk (*) denotes freezing the classifier dimensions corresponding to historical tasks. Comparing (1-1) and (1-2) reveals that unconstrained classifier updates are a primary forgetting source; freezing historical weights yields substantial gains, which are further enhanced by the cosine classifier (1-3) through decoupling feature norms from directional alignment. Introducing basic partitioning (2-1) improves stability, but Group-Aware Parameter Partitioning (GAPP, 2-2) delivers a dramatic performance leap (+15.4% Acc on ImageNet-R), validating that magnitude-guided, structure-aware isolation effectively prevents parameter overwriting within shared experts. Adding orthogonal gradient projection (2-3) consistently improves results by constraining updates to the null space of historical feature directions. The dynamic capacity fallback (3-1) proves essential for long streams, automatically preventing representational saturation when task complexity increases. Finally, integrating the full hierarchical moment regularization $\mathcal{L}_{\text{reg}}$ (3-3) achieves peak performance, confirming that jointly tracking intra-expert ($\mathcal{M}^{\text{cur}}$) and inter-expert ($\mathcal{M}^{\text{hist}}$) covariance structures is necessary for long-term stability. Collectively, each component addresses a distinct interference pathway, operating synergistically to maximize parameter efficiency without catastrophic forgetting.

Importance Metric and Cost. Tab. 4 isolates the choice of importance metric within GAPP. Random masking collapses on long streams (45.49% Acc on ImageNet-R 20 tasks), confirming that structure-aware allocation is essential. Diagonal Fisher and our parameter magnitude are broadly comparable:

Table 5: Sensitivity to the saturation threshold β. Each cell reports the approximate boundary β, the number of experts, Acc (%), and the average covariance memory per task (MiB), written as β (experts, Acc, MiB). One full covariance matrix requires 2.25 MiB in float32. Columns are ordered from fewer to more capacity fallbacks (i.e., decreasing β) relative to the default $\beta = 1$, *Worse* and *Better* correspond to skipping and adding one capacity fallback, respectively.

Dataset	Fewer fallbacks		Default ($\beta = 1$)	More fallbacks	
	No fallback ($\beta \geq$)	Worse ($\beta \geq$)		Better ($\beta \leq$)	All fallback ($\beta \leq$)
INR (5)	–	1.20 (3, 77.18, 1.35)	1.00 (4, 85.89, 1.80)	–	–
INR (10)	1.39 (4, 79.35, 0.90)	1.09 (5, 83.69, 1.13)	1.00 (6, 85.45, 1.35)	0.94 (6, 86.01, 1.35)	0.82 (7, 86.87, 1.58)
INR (20)	–	–	1.00 (6, 82.76, 0.68)	0.98 (7, 83.39, 0.79)	0.73 (14, 90.29, 1.58)
INA (10)	–	1.19 (4, 60.26, 0.90)	1.00 (5, 65.84, 1.13)	0.97 (6, 66.02, 1.35)	0.89 (7, 66.45, 1.58)
DN (5)	–	–	1.00 (3, 73.10, 1.35)	0.97 (4, 76.32, 1.80)	–

Table 6: Comparison on ImageNet-R (20 tasks) in terms of parameter efficiency.

Method	Expert No.	Total Parameters (M)	Learnable Parameters (M)	GFLOPs
Hide-Prompt	20	88.17	0.08	70.36
SD-LoRA	20	93.97	0.37	35.12
LoRA-P&M	20	93.97	0.37	35.12
$\mathbf{E}^3$	6	88.79	$\sim$0.11	35.12

Fisher leads on six of eight metrics, while magnitude is better on ImageNet-R 5-task Acc (+0.83%) and ImageNet-A Acc (+1.05%). But this comes at a substantial cost: storing a diagonal Fisher matrix for each LoRA expert requires about 1.5 GB of additional memory and 0.2 h of extra computation on ImageNet-R 10-task, whereas magnitude is read directly from the adapted weights without extra backward passes. We therefore view magnitude as a near-parity, lower-cost alternative to Fisher rather than as a uniformly stronger metric.

Sensitivity to the Saturation Threshold. Tab. 5 traces how β trades expert number for efficiency. Increasing β suppresses capacity fallback: on ImageNet-R (10 tasks), $\beta \geq 1.39$ eliminates fallback, reducing the expert count from 6 to 4 but lowering Acc from 85.45% to 79.35%. Decreasing β triggers additional fallback: $\beta \leq 0.82$ reaches 86.87% Acc with 7 experts, while on ImageNet-R (20 tasks) $\beta \leq 0.73$ expands to 14 experts and raises Acc from 82.76% to 90.29%. Similar behavior appears on DomainNet (73.10% with 3 experts at $\beta = 1$ vs. 76.32% with 4 experts at $\beta \leq 0.97$). Hence $\beta = 1$ is an efficiency-oriented operating point rather than an accuracy ceiling. It stays within the stable range on all benchmarks (e.g., $[0.94, 1.09]$ on ImageNet-R 10-task and $[0.97, 1.19]$ on ImageNet-A) while keeping the average covariance memory at an acceptable level. Moving to a higher-accuracy regime raises this cost (e.g., DomainNet: 1.35 $\rightarrow$ 1.80 MiB/task). The 5-task ImageNet-R stream is an exception as $\beta = 1$ already operates in an all-fallback regime.

Analysis of Parameter Efficiency. As shown in Tab. 6, $\mathbf{E}^3$ also demonstrates superior parameter and computational efficiency. $\mathbf{E}^3$ completes the 20-task sequence with only 6 experts, significantly reducing structural redundancy. In terms of computational cost, $\mathbf{E}^3$ requires only 35.12 GFLOPs per expert, half the GFLOPs of Hide-Prompt, while also maintaining fewer total (88.79M) and learnable ($\sim$0.11M) parameters than SD-LoRA and LoRA-P&M. These results confirm that $\mathbf{E}^3$ effectively balances model capacity, parameter efficiency, and inference cost through its elastic expert allocation.

5 Conclusion and Limitations

This paper presents $\mathbf{E}^3$, an elastic ensemble of LoRA experts for parameter-efficient class-incremental learning. $\mathbf{E}^3$ mitigates forgetting while improving efficiency through dynamic expert capacity scheduling, group-aware parameter partitioning, and dual-level interference mitigation, without rehearsal, inference-time task identifiers, or costly gradient-sensitivity analysis. Experiments show that $\mathbf{E}^3$ achieves state-of-the-art or competitive performance with compact memory and sublinear inference-latency growth. Limitations remain in capacity scheduling, which lacks theoretical guarantees and leaves room for refinement; test-time expert selection, which evaluates all experts and is thus inefficient; and covariance-statistic storage, which incurs memory overhead. Future work will therefore strengthen the theoretical analysis and refinement of capacity scheduling, develop more efficient test-time expert selection, and reduce the memory cost of stored statistics.

6 Acknowledgments

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Table 7: Performance comparison on CIFAR-100 and CUB-200.

Method	CIFAR-100 10 tasks		CUB-200 10 tasks	
	Acc ↑	AAA ↑	Acc ↑	AAA ↑
Full Fine-Tuning	$69.49_{(0.50)}$	$80.35_{(0.87)}$	$51.53_{(1.41)}$	$69.74_{(0.93)}$
L2P [26]	$83.18_{(1.20)}$	$87.69_{(1.05)}$	$65.18_{(2.49)}$	$76.12_{(1.27)}$
DualPrompt [25]	$81.48_{(0.86)}$	$86.41_{(0.66)}$	$68.00_{(1.06)}$	$79.40_{(0.88)}$
CODA-Prompt [19]	$86.31_{(0.12)}$	$90.67_{(0.22)}$	$71.92_{(0.33)}$	$78.76_{(0.65)}$
InfLoRA [14]	$86.75_{(0.35)}$	$91.72_{(0.15)}$	$70.82_{(0.23)}$	$81.39_{(0.14)}$
SD-LoRA [27]	$88.01_{(0.31)}$	$92.54_{(0.18)}$	$77.48_{(0.20)}$	$85.59_{(0.44)}$
LoRA-P&M [17]	$88.45_{(0.35)}$	$92.89_{(1.13)}$	$78.29_{(0.50)}$	$83.39_{(0.61)}$
$\mathbf{E}^3$	$\underline{90.41}_{(0.13)}$	$\underline{94.72}_{(0.35)}$	$\underline{83.81}_{(0.24)}$	$\underline{85.64}_{(0.60)}$
$\mathbf{E}^3$ (Full Expansion)	$\mathbf{96.47}_{(0.39)}$	$\mathbf{97.13}_{(0.46)}$	$\mathbf{86.79}_{(0.18)}$	$\mathbf{87.25}_{(0.26)}$

A Additional Implement Details

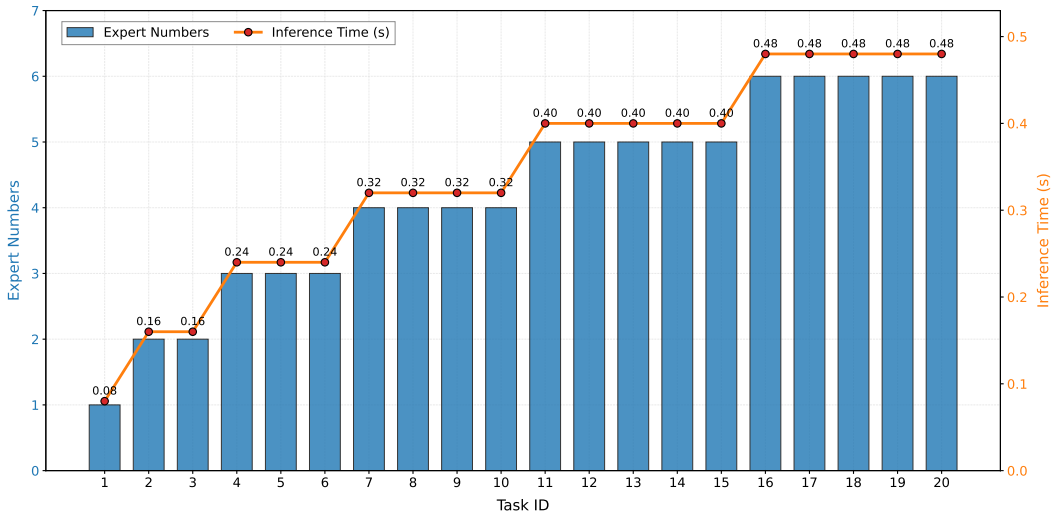


Figure 3: Evolution of expert count and inference time during training on ImageNet-R (20 Tasks).

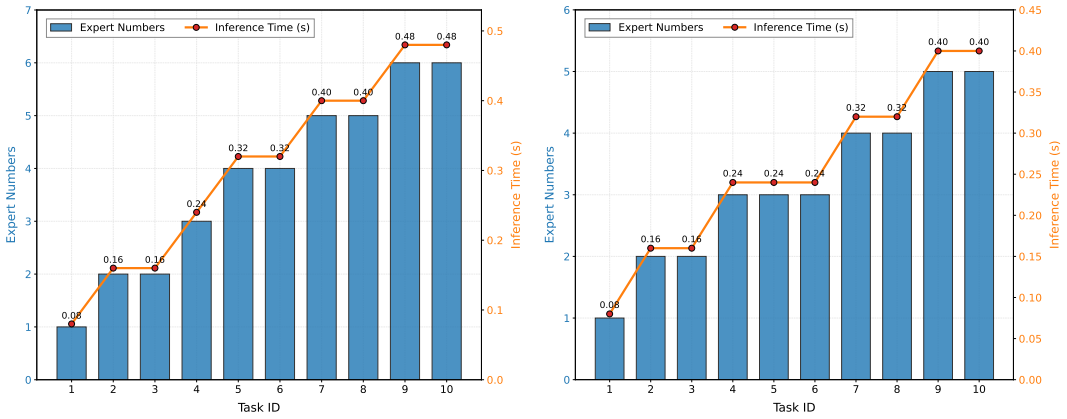
The projection strength α in Eq. (18), the loss weight λ_{reg} in Eq. (20) and cross-response threshold β in Eq. (22) are set to 0.5, 1, and 1, respectively. The value $\lambda_{\text{reg}} = 1$ is chosen heuristically, based on the observation that its corresponding loss term is comparable in magnitude to the cross-entropy loss, and that model performance remains largely stable under moderate variations of λ_{reg} around this value. In addition, the σ in stage 1 in Section 3.3 is set to 5 (half of the total 10 epochs) for all datasets. The top-d in $\mathbf{Q}_i \in \mathbb{R}^{D\times d}$ is set to 16. All methods compared in the experiments employ identical training configurations as detailed in Section 4.1.

The overall algorithmic workflow of this paper is delineated in Algorithm 1.

B Expert Selection During Testing

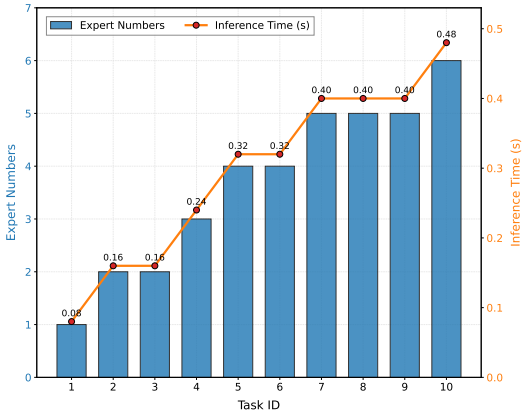
In the strict class-incremental learning (CIL) setting, task identifiers are unavailable during inference. Consequently, the model must autonomously route each query to the most appropriate expert $\mathcal{E}_k \in \mathcal{E}$ for prediction. We adopt a lightweight, self-supervised routing strategy based on **predictive entropy minimization**. The effectiveness of this strategy has been validated in recent works [23].

Let $\mathcal{E} = \{\mathcal{E}_1, \ldots, \mathcal{E}_K\}$ denote the set of existing experts accumulated after training on N tasks. For a given input batch $\mathcal{B} = \{\mathbf{x}_i\}_{i=1}^{|\mathcal{B}|}$, we independently extract features and compute softmax probability distribution $\mathbf{p}_{k,i}$ for each expert. We then calculate the batch-averaged predictive entropy for the k-th

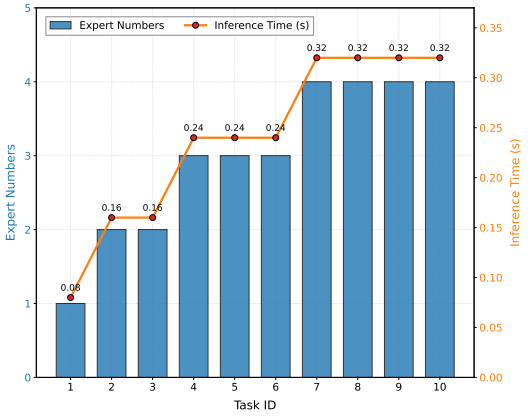


(a) Evolution on ImageNet-R (10 tasks).

(b) Evolution on ImageNet-A (10 tasks).



(c) Evolution on CIFAR-100 (10 tasks).



(d) Evolution on CUB-200 (10 tasks).

Figure 4: Evolution of expert count and inference time during training on other 10-task sequences.

expert:

$$H_k(\mathcal{B}) = \frac{1}{|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \left(-\sum_{c=1}^{C} p_{k,i,c} \log(p_{k,i,c} + \epsilon) \right), \tag{23}$$

where C is the cumulative number of classes, and $\epsilon = 10^{-10}$ ensures numerical stability against zero probabilities. The selected expert index $\hat{k}$ is determined by:

$$\hat{k} = \arg \min_{k \in \{1,\ldots,K\}} H_k(\mathcal{B}). \tag{24}$$

Intuitively, a lower predictive entropy indicates higher output confidence, implying that the batch $\mathcal{B}$ aligns well with the feature manifold and decision boundaries specialized by expert $\mathcal{E}_{\hat{k}}$. This criterion naturally routes samples to their corresponding experts without requiring auxiliary routers, task metadata, or stored exemplars.

Additionally, we quantify the overall inference latency (expert selection & forward pass latency per test sample) on our current hardware setup. As presented in Fig. 3, thanks to our elastic expert strategy, the inference latency scales sub-linearly with the number of tasks.

C Additional Results and Analysis

In addition to ImageNet-R, ImageNet-A, and DomainNet, we evaluate the proposed $\mathbf{E}^3$ on two other widely used benchmarks: CIFAR-100 [12] and CUB-200 [20]. The results show that $\mathbf{E}^3$ outperforms all other baseline methods on both datasets, thereby further demonstrating the superiority of our approach.

As shown in Fig. 4 and 5, $\mathbf{E}^3$ efficiently creates a suitable number of experts across the remaining sequences. Specifically, expert saturation is detected in task 3 (Fig. 4a) and task 8 (Fig. 4b). In the

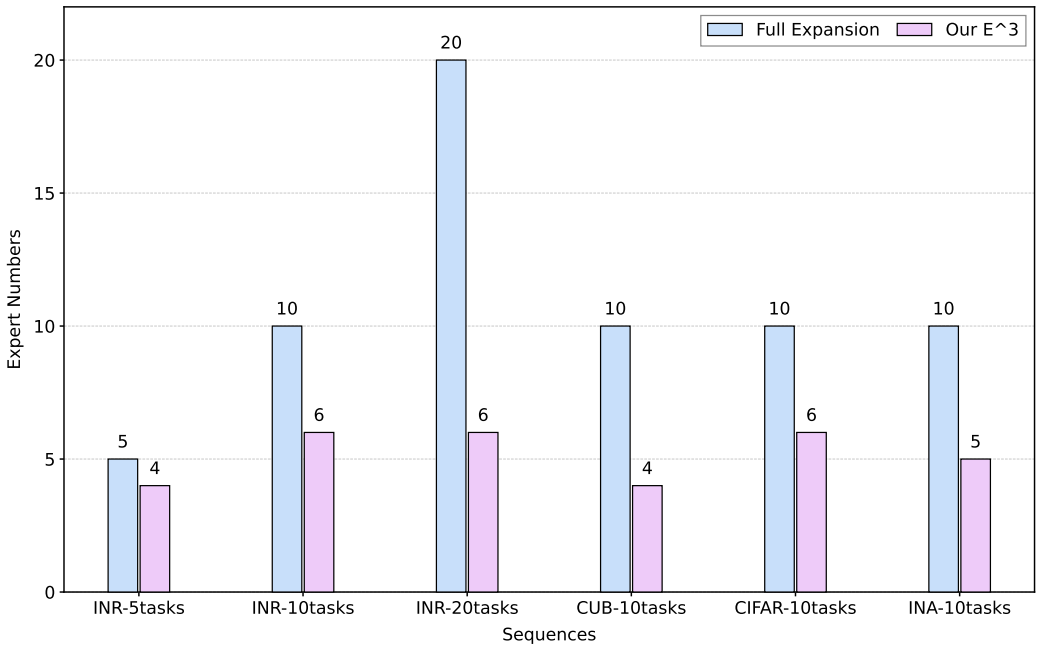
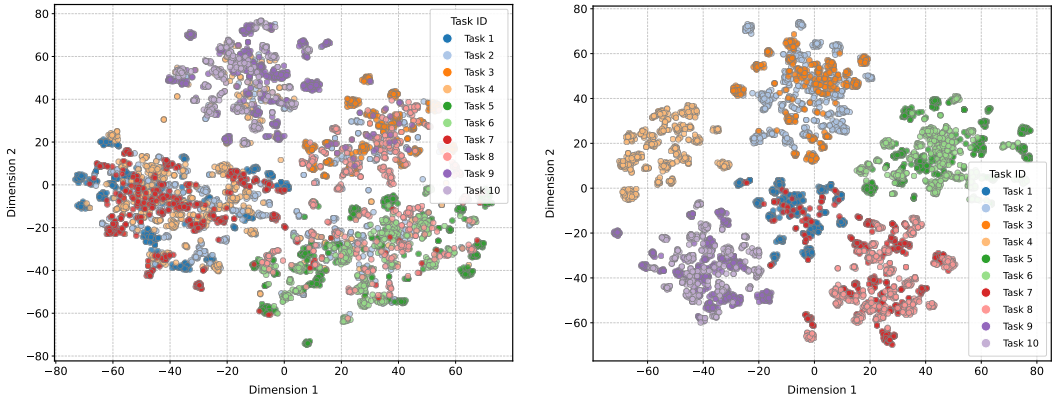


Figure 5: Comparison of Expert Numbers: Full Expansion vs. $\mathbf{E}^3$.



(a) $\mathbf{E}^3$ w/o $\mathcal{L}_{\text{reg}}$. (b) $\mathbf{E}^3$ w/ $\mathcal{L}_{\text{reg}}$.

Figure 6: Comparison of feature distributions extracted by individual experts for their corresponding tasks after training on all tasks of ImageNet-R (10 tasks).

former case, saturation occurs just as the expert with capacity "2" finishes its last assigned task, after which a new expert with capacity "1" is instantiated. In the latter, saturation is triggered early (after only the second task assigned to an expert of capacity "4"), prompting immediate termination of that expert, and the creation of a replacement with capacity "3" to learn the remaining tasks.

Furthermore, as shown in Fig. 6, the features extracted by each expert with $\mathcal{L}_{\text{reg}}$ exhibit significantly improved separability compared to those without it. This enhanced feature discrimination, coupled with the performance gains shown in Tab. 3, indicates that $\mathcal{L}_{\text{reg}}$ acts as a geometric regularizer. It simultaneously anchors historical knowledge and steers new task adaptation towards orthogonal subspaces, thereby preventing catastrophic drift in established features while reserving complementary capacity for future tasks.

Algorithm 1 Training Procedure for $\mathbf{E}^3$

Require: Task stream $\{\mathcal{T}_t\}_{t=1}^{N}$, pre-trained backbone F_Ω, hyperparameters $\{\sigma, E_{\max}, \lambda_{\text{reg}}, \alpha, d\}$
Ensure: Expert sequence $\mathcal{E}$ and classifier $\mathbf{W}$

Initialize $\mathcal{E}_1, \mathcal{C}_1 \leftarrow 1$, $\mathbb{M}^{\text{freeze}} \leftarrow \mathbf{1}$, $\mathcal{M}^{\text{cur}} \leftarrow \emptyset$, $\mathcal{M}^{\text{hist}} \leftarrow \emptyset$
Set current expert index $k \leftarrow 1$
for $t = 1$ **to** N **do**
  if $t > t_k^{\text{start}} + \mathcal{C}_k - 1$ **or** saturation via Eq. (22) **then**
    Archive and freeze $\mathcal{E}_k$
    $\mathbf{M}_k^{\text{hist}} \leftarrow \text{Avg}(\mathcal{M}^{\text{cur}})$ via Eq. (14)
    $\mathcal{M}^{\text{hist}} \leftarrow \mathcal{M}^{\text{hist}} \cup \{\mathbf{M}_k^{\text{hist}}\}$
    if saturation triggered **then**
      $\mathcal{C}_{k+1} \leftarrow \max(\mathcal{C}_k - 1, 1)$ {Eq. (4)}
    else
      $\mathcal{C}_{k+1} \leftarrow \mathcal{C}_k + 1$ {Eq. (3)}
    end if
    $k \leftarrow k + 1$, instantiate $\mathcal{E}_k$, reset $\mathbb{M}^{\text{freeze}} \leftarrow \mathbf{1}$, $\mathcal{M}^{\text{cur}} \leftarrow \emptyset$, $t_k^{\text{start}} \leftarrow t$
  end if
  {**Two-Stage Optimization for** $\mathcal{T}_t$}
  Identify $\Theta_{\text{free}} \subseteq \Theta_{\text{LoRA}}^{(k)}$
  for $e = 1$ **to** $E_{\max}$ **do**
    Compute $\mathcal{L}_{\text{total}}$ via Eq. (20)
    Backward pass $\Rightarrow$ raw gradients $\{\nabla\theta\}$
    if $e \leq \sigma$ **then**
      $\nabla\theta \leftarrow \nabla\theta \odot \mathbb{M}^{\text{freeze}}$ {Preliminary probing}
    else if $e = \sigma + 1$ **then**
      Compute s^{masked} via Eq. (9), threshold via Eq. (6)
      Generate $\mathbb{M}^{(t)}$ via Eq. (7), update $\mathbb{M}^{\text{freeze}}$ via Eq. (8)
      $\nabla\theta \leftarrow \nabla\theta \odot \mathbb{M}^{(t)}$
    else
      $\nabla\theta \leftarrow \nabla\theta \odot \mathbb{M}^{(t)}$ {Constrained training}
    end if
    Compute $\mathbf{P}_k$ via Eq. (18) using $\mathbf{M}_k^{\text{ref}} = \text{Avg}(\mathcal{M}^{\text{cur}})$
    Project gradients $\mathbf{G}_k^{\text{proj}}$ via Eq. (19)
    Optimizer step with $\mathbf{G}_k^{\text{proj}}$
  end for
  {**End-of-Task Update**}
  Extract features from $\mathcal{D}_t$, compute $\mathbf{M}_t$ via Eq. (11)
  $\mathcal{M}^{\text{cur}} \leftarrow \mathcal{M}^{\text{cur}} \cup \{\mathbf{M}_t\}$
  Evaluate saturation condition via Eq. (21)–(22)
end for
return $\mathcal{E}$, $\mathbf{W}$

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